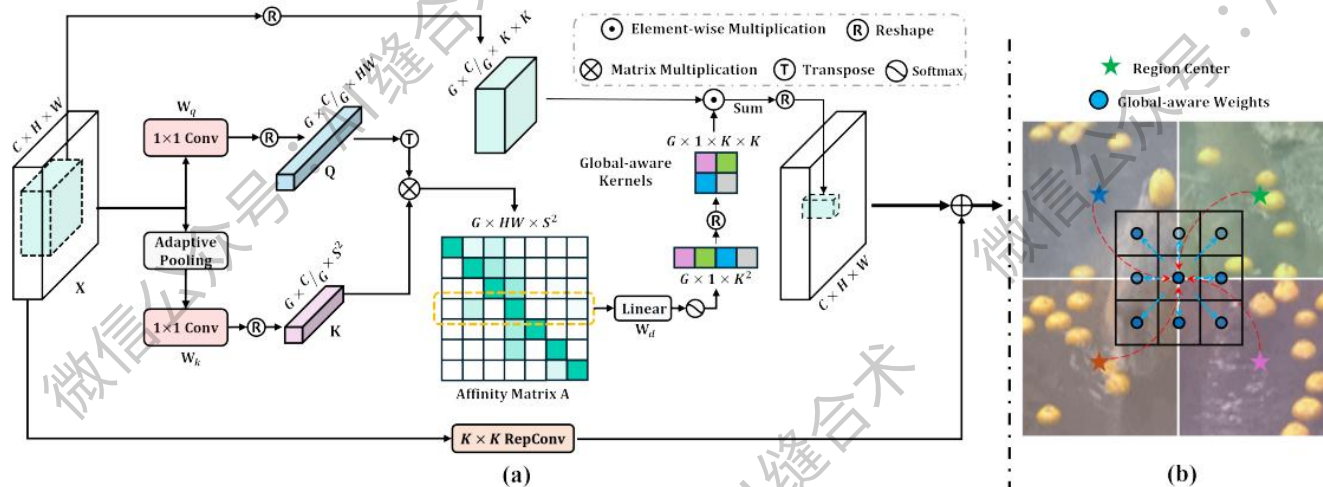


核心速览



论文题目: OverLoCK: An Overview-first-Look-Closely-next ConvNet with Context-Mixing Dynamic Kernels

中文题目: OverLoCK: 一种先概览-再仔细观察-最后上下文混合动态核的卷积网络

论文链接: <https://arxiv.org/pdf/2502.20087>

官方 github: <https://github.com/LMMMEng/OverLoCK/tree/main>

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核心速览: 本文提出了一种名为 OverLoCK 的新型卷积神经网络 (ConvNet) 架构, 该架构首次在纯 ConvNet 中明确集成了自顶向下的注意力机制, 通过上下文混合动态核 (ContMix) 技术有效模拟长距离依赖关系, 同时保留了局部归纳偏置。在图像分类、目标检测、实例分割和语义分

割等视觉任务上均超越众多基于 CNN、Transformer 和 Mamba 的骨干网络，并且在速度和准确性权衡方面具有显著优越性！

上下文混合动态卷积 (ContMix)： ContMix 是一种新颖的动态卷积方法，它能够动态地模拟长距离依赖关系，同时保持强烈的局部归纳偏置。通过计算输入特征图中每个 token 与一组区域中心的亲和度，生成空间变化的动态卷积核，使得每个卷积核权重携带来自上下文的全球信息。

```
ContMix(
  (lepe): Sequential(
    (0): DilatedReparamBlock(
      (lk_origin): Conv2d(64, 64, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), groups=64, bias=False)
      (origin_bn): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (dil_conv_k5_1): Conv2d(64, 64, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=64, bias=False)
      (dil_bn_k5_1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (dil_conv_k3_1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=64, bias=False)
      (dil_bn_k3_1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (dil_conv_k3_2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(2, 2), dilation=(2, 2), groups=64, bias=False)
      (dil_bn_k3_2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (dil_conv_k3_3): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(3, 3), dilation=(3, 3), groups=64, bias=False)
      (dil_bn_k3_3): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    )
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
  (weight_query): Sequential(
    (0): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
  (weight_key): Sequential(
    (0): AdaptiveAvgPool2d(output_size=7)
    (1): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (2): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
  (weight_proj): Conv2d(49, 74, kernel_size=(1, 1), stride=(1, 1))
  (dyconv_proj): Sequential(
    (0): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
)
Input shape: torch.Size([1, 64, 32, 32])
Output shape: torch.Size([1, 64, 32, 32])
```